TECHNOFuture KAFR EL SHEIKH | AFFIAT · ASWAN FINALS 2026

GAMES CATEGORY · 2026

Pixel Verse Studios

Real science, hidden in play.

Two 16-bit games for ages 5 to 99. Dodge ocean trash. Fly to Jupiter's icy moon. Every fact in them is real.

01 · WHY Facts stick when you play them.

THE PROBLEM

A fact in a lesson stays a sentence

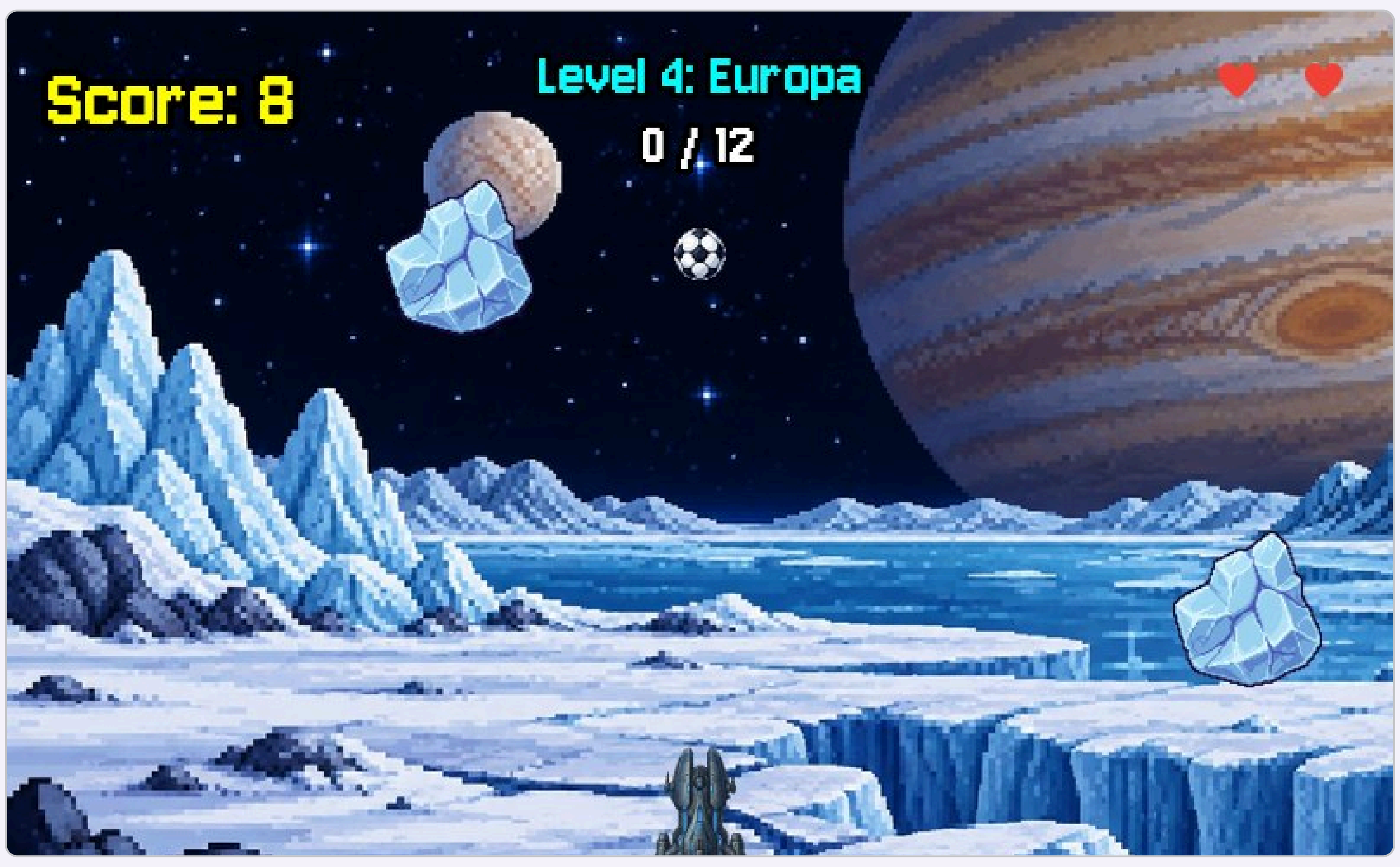
"Europa hides an ocean." "Plastic hurts sea turtles." A child hears it, and the next day it is gone. Facts feel far away from play.

OUR SOLUTION

Make the fact a game moment

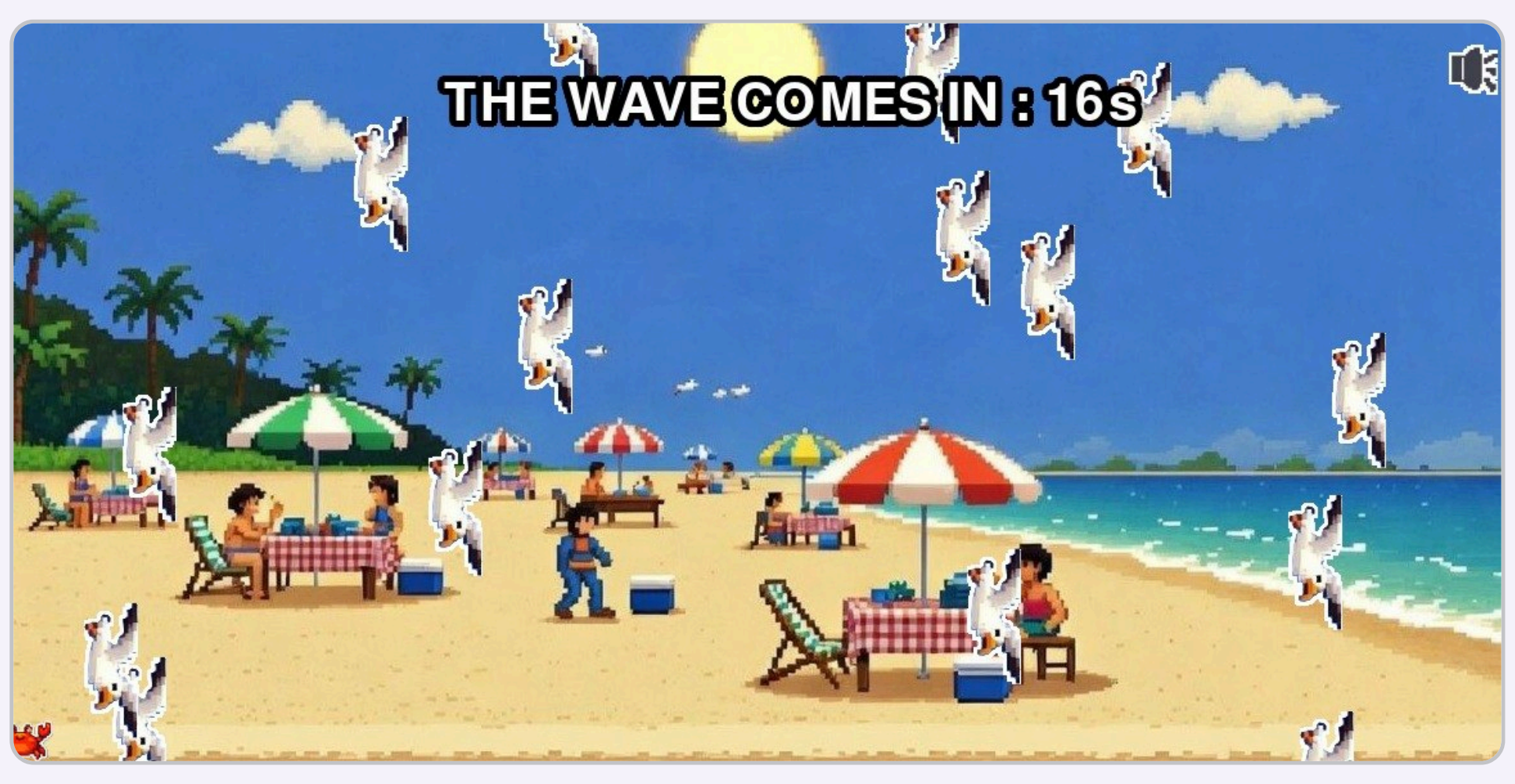
In Crab Quest you dodge the trash. In Space X you fly through Europa's ice. Each fact shows up where it makes sense in the story, and nothing in it is invented.

02 · SEE IT Two games, ten levels.

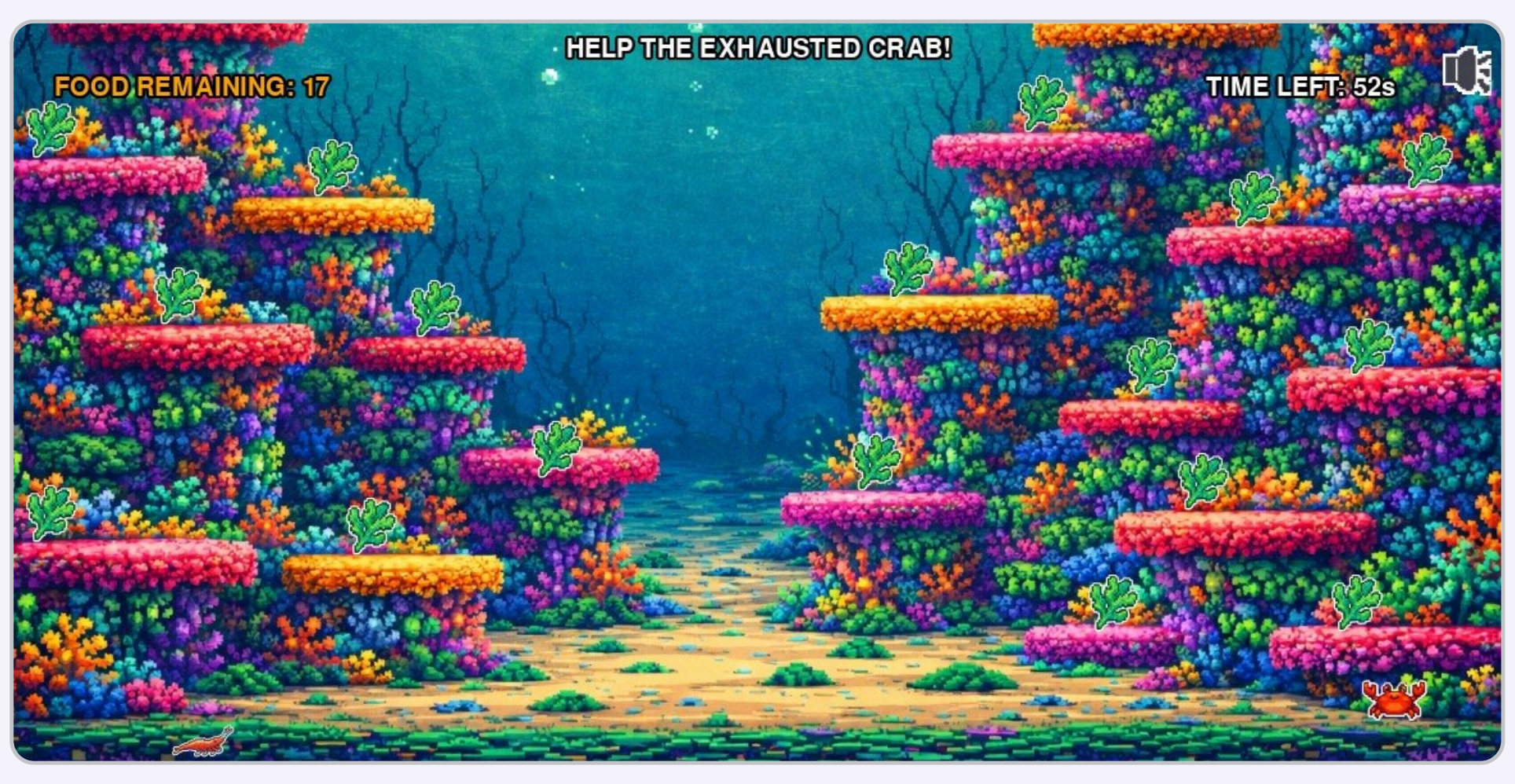


Space X, level 4: the Ship breaks Europa's ice with footballs. Each chunk needs two hits, and the Europa level ends with a 3-second scan.

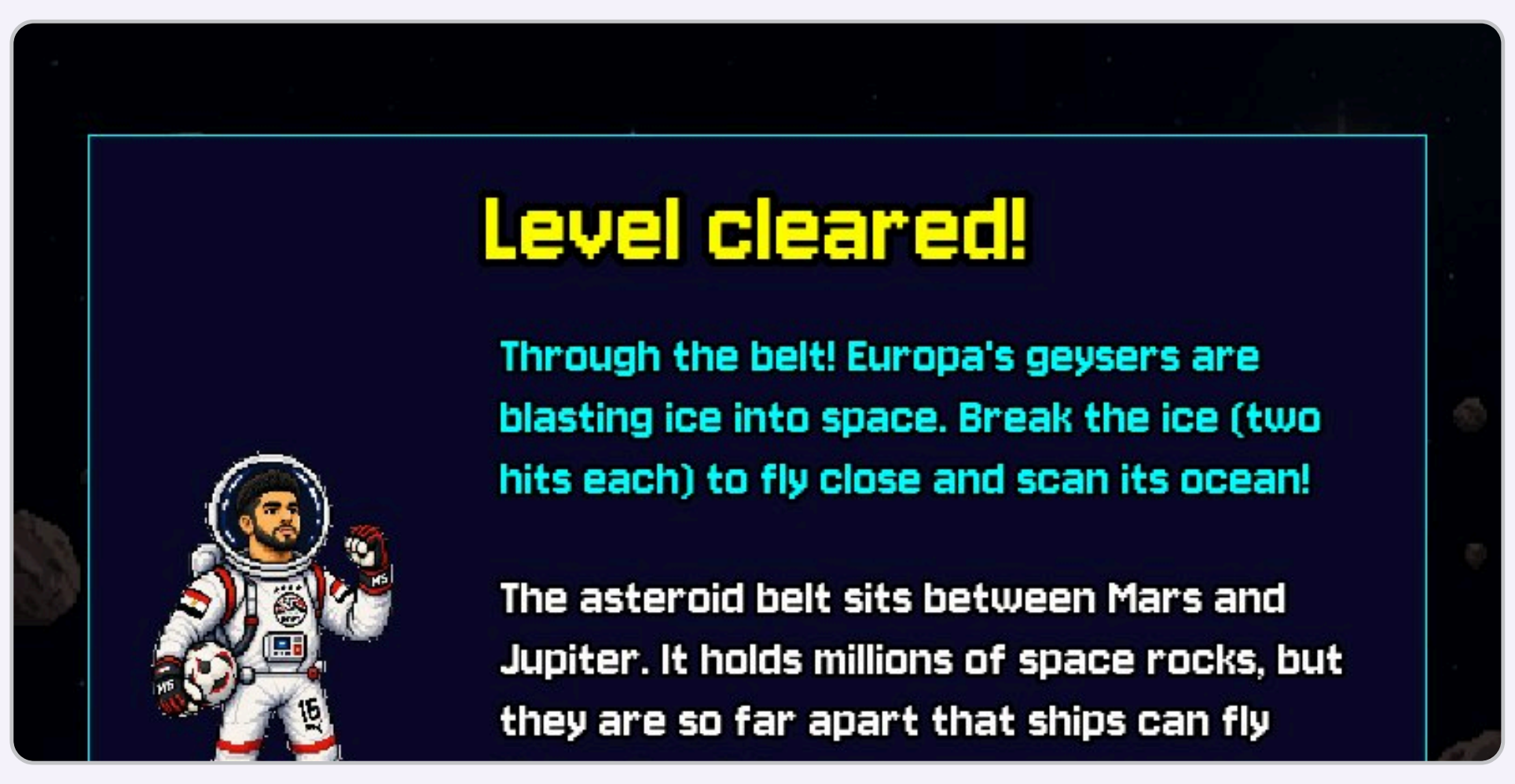
- **Crab Quest: 6 chapters**
  Platforming, dodging, collecting and surviving, from a kitchen to a coral sanctuary.
- **Space X: 4 levels**
  Launch pad, Earth orbit, asteroid belt and Europa, then the win screen on Jupiter.
- **A fact card after every level**
  A story line and one science fact in Space X.
- **A planet fact on every game over**
  The Sun and all eight planets, each with a picture.
- **Keyboard or on-screen buttons**
  Arrows, Space, P and R, plus clickable left, right and pause in Space X.
- **Music, mute and fullscreen**
  A mute button in Crab Quest; pause and fullscreen (F) in Space X.



Crab Quest 2: Beach Run, dodge the birds.



Crab Quest 4: Shrimp Help, collect 17 algae.



Level cleared!

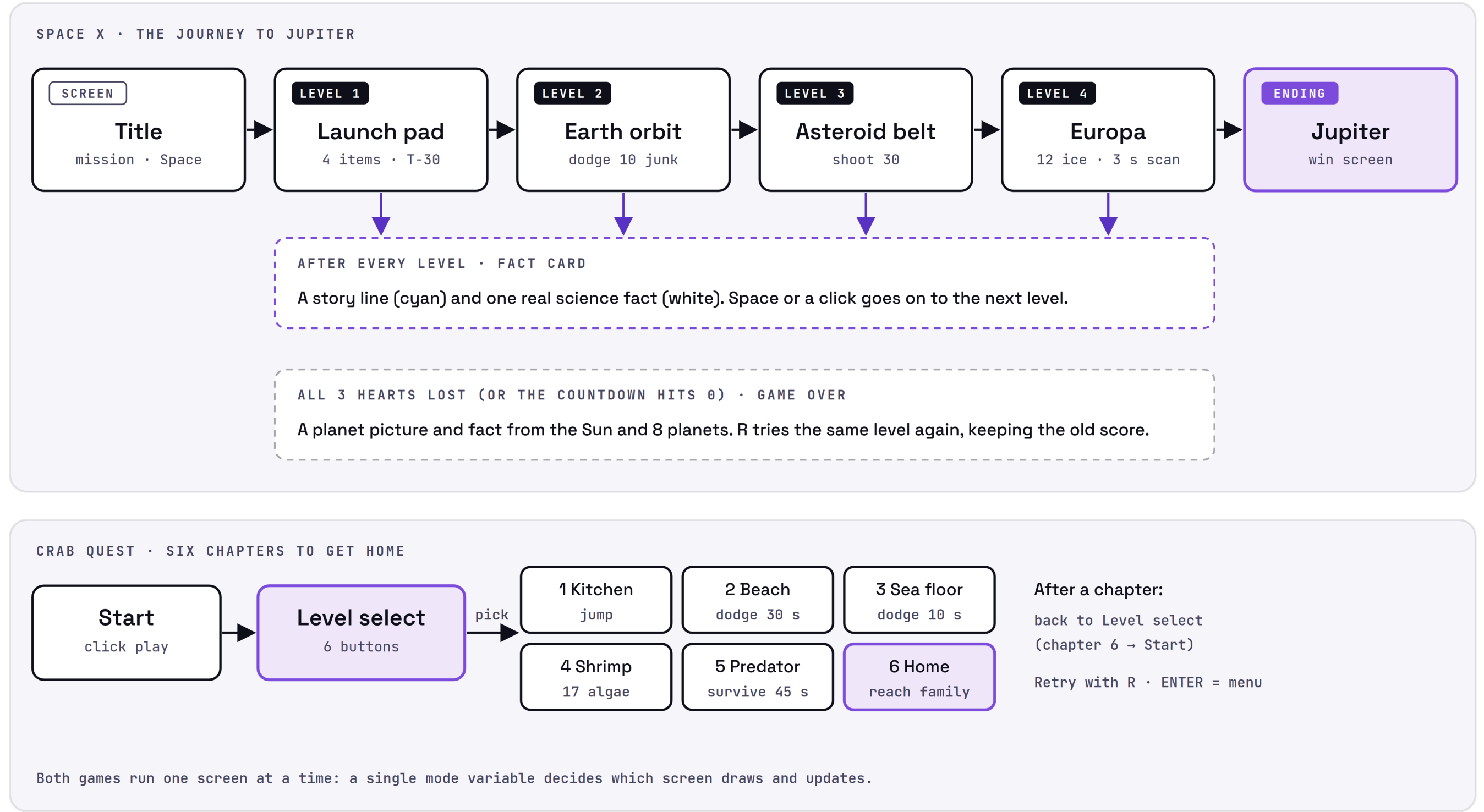
Through the belt! Europa's geysers are blasting ice into space. Break the ice (two hits each) to fly close and scan its ocean!

The asteroid belt sits between Mars and Jupiter. It holds millions of space rocks, but they are so far apart that ships can fly

Space X: the fact card after a level.

03 · HOW One mode switch, plain Python.

HOW A PLAYER MOVES THROUGH THE GAMES



SPACE X · THE JOURNEY TO JUPITER

SCREEN Title — mission · Space → LEVEL 1 Launch pad — 4 items · T-30 → LEVEL 2 Earth orbit — dodge 10 junk → LEVEL 3 Asteroid belt — shoot 30 → LEVEL 4 Europa — 12 ice · 3 s scan → ENDING Jupiter — win screen

AFTER EVERY LEVEL · FACT CARD
A story line (cyan) and one real science fact (white). Space or a click goes on to the next level.

ALL 3 HEARTS LOST (OR THE COUNTDOWN HITS 0) · GAME OVER
A planet picture and fact from the Sun and 8 planets. R tries the same level again, keeping the old score.

CRAB QUEST · SIX CHAPTERS TO GET HOME

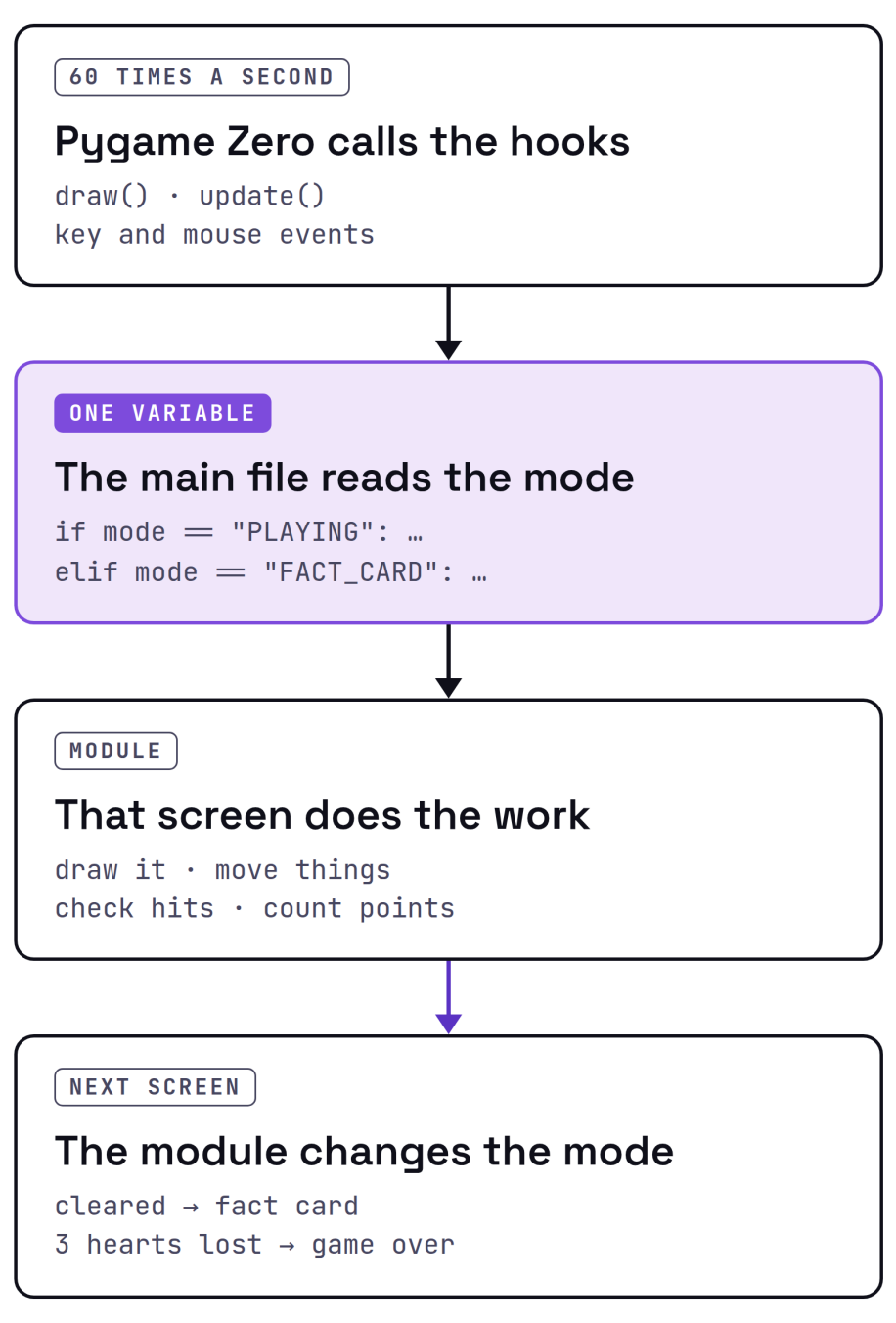
Start — click play → Level select — 6 buttons → pick → 1 Kitchen — jump · 2 Beach — dodge 30 s · 3 Sea floor — dodge 10 s · 4 Shrimp — 17 algae · 5 Predator — survive 45 s · 6 Home — reach family

After a chapter:
back to Level select (chapter 6 → Start)
Retry with R · ENTER = menu

Both games run one screen at a time: a single mode variable decides which screen draws and updates.

HOW BOTH GAMES ARE BUILT

HOW A FRAME IS DRAWN · BOTH GAMES



60 TIMES A SECOND
Pygame Zero calls the hooks
`draw() · update()`
key and mouse events

ONE VARIABLE
The main file reads the mode
`if mode == "PLAYING": …`
`elif mode == "FACT_CARD": …`

MODULE
That screen does the work
draw it · move things
check hits · count points

NEXT SCREEN
The module changes the mode
cleared → fact card
3 hearts lost → game over

WHO OWNS WHAT

	CRAB QUEST	SPACE X
ENTRY FILE	`main.py`	`Game.py`
MODE VARIABLE	`game_states.game_mode`	`parts/game_states.mode`
MODES	START_SCREEN, LEVEL_SELECT, LEVEL1 to LEVEL6	TITLE, PLAYING, FACT_CARD, GAME_OVER, WIN
LEVELS	`level_1.py` to `level_6.py`	`levels/`: launch_pad, earth_orbit, belt, europa
OTHER SCREENS	`start_screen.py`, `level_select.py`	`screens/`: title, fact_card, game_over, win
SHARED PARTS	Mute button in `main.py`	`parts/`: ship, controls, hud
FACTS & NUMBERS	Inside each level module	`data/`: level_data, game_data
ASSETS	`images/` · `music/`	`images/` · `music/` · `fonts/`
PYTHON FILES	10	19

No classes in either game. Each module can be read on its own, top to bottom.

2
games: Crab Quest and Space X

10
levels: 6 chapters and 4 flights

29
Python files across both games

0
classes: plain variables, ifs and loops

BUILT WITH

Python | Pygame Zero | Pygame | Pixel art | HTML + CSS

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PRESENTED AT
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Scan to visit the studio
`ibrhmahmd.github.io/PVS`

AFFIAT · Afro-Asian Forum for Innovation and Technology · Aswan Finals 2026 · Games category

Every fact in both games is real. All screenshots are from the running games.